
Contents

Editor Biography	x
Contributors.....	xii
Preface.....	xv
Acknowledgments.....	xvii

Chapter 1	Leveraging IoT for Precision Health Monitoring in Livestock with Artificial Intelligence	1
	<i>Devinder Kaur and Amandeep Kaur Virk</i>	

Chapter 2	Significance of Machine Learning in Apple Disease Detection and Implications.....	19
	<i>Saimul Bashir, Syed Nisar Hussain Bukhari, Faisal Firdous, and Gursimran Jeet Kour</i>	

Chapter 3	Intelligent Inputs Revolutionizing Agriculture: An Analytical Study.....	42
	<i>Nelofar Ara, Sani Mustapha Kura, VK Aswathy, and Mohammad Amin Wani</i>	

Chapter 4	Case Studies on the Initiatives and Success Stories of Edge AI Systems for Agriculture.....	56
	<i>Thamizhiniyan Natarajan and Shanmugavadivu Pichai</i>	

Chapter 5	Crop Recommender: Machine Learning–Based Computational Method to Recommend the Best Crop Using Soil and Environmental Features	81
	<i>Syed Nisar Hussain Bukhari, Jewiara Khursheed Wani, Ummer Iqbal, and Muneer Ahmad Dar</i>	

Chapter 6	A Perusal of Machine-Learning Algorithms in Crop-Yield Predictions	101
	<i>Anshika Gupta, Mohit Soni, and Kalpana Katiyar</i>	

Chapter 7	Harvesting Intelligence: AI and ML Revolutionizing Agriculture	126
	<i>Arya Kumari, Muhammad Najeeb Khan, and Amit Kumar Sinha</i>	
Chapter 8	Using Deep Learning to Detect Apple Leaf Disease	142
	<i>Syed Nisar Hussain Bukhari, Rukaya Manzoor, Ummer Iqbal, and Muneer Ahmad Dar</i>	
Chapter 9	Agricultural Crop-Yield Prediction: Comparative Analysis Using Machine Learning Models.....	158
	<i>Kukatlapalli Pradeep Kumar, S. Babu Kumar, Amarthya Dutta Gupta, Kevin Johnson, and Meghan Mary Michael</i>	
Chapter 10	Fundamentals of AI and Machine Learning with Specific Examples of Application in Agriculture.....	178
	<i>Manoj Kumar Mahto, P. Laxmikanth, and V.S.S.P.L.N. Balaji Lanka</i>	
Chapter 11	Farming Futures: Leveraging Machine Language for Potato Leaf Disease Forecasting and Yield Optimization.....	200
	<i>A. Vijayalakshmi, A. Nidin, and Deepthi Das</i>	
Chapter 12	Classification of Farms for Recommendation of Rice Cultivation Using Naive Bayes and SVM: A Case Study	216
	<i>Qurat-ul-ain, Uzma Hameed, and Hamira Mehraj</i>	
Chapter 13	Neural Networks for Crop Disease Detection.....	227
	<i>Mohammad Ubaidullah Bokhari, Gaurav Yadav, and Md. Zeyauddin</i>	
Chapter 14	Short-Term Weather Forecasting for Precision Agriculture in Jammu and Kashmir: A Deep-Learning Approach	243
	<i>Syed Nisar Hussain Bukhari and Sana Farooq Pandit</i>	

Chapter 15	Deep Reinforcement Learning for Smart Irrigation	259
-------------------	--	-----

Aakansha Khanna and Inzimam Ul Hassan

Index	280
-------------	-----